



INDIA'S ENERGY RESILIENCE COMMAND CENTER

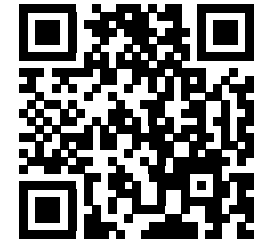
FINAL PROJECT DOCUMENT

Sanjiv

A command center for the moment a supply shock stops being a headline and becomes a decision. It shows what is known, what is assumed, what happens next, and who must approve the response.

WHY SANJIV EXISTS

Energy resilience is not one prediction. It is a chain of imperfect signals, physical constraints and human responsibility. I built Sanjiv to keep that chain visible - especially the uncertainty.



SCAN FOR REPOSITORY

[GitHub repository](#)
[Demo folder](#)
vivekyarra567@gmail.com

Built and documented by Yarra Vivek

Submission snapshot | 2026-07-22

Commit
14a7e5deb639aeb6de4e683ec62cd22abbea0838

All screenshots and measured values are traceable to repository artifacts. PortWatch is a live-fetched daily observation; fixture and replay evidence is never presented as live vessel tracking.

02 / PROBLEM

A disruption does not arrive with clean data

The team still has to answer: What changed? What breaks next? What can we do? Which numbers are trustworthy? Who signs off?

IMAGINE THE MOMENT

The Strait of Hormuz loses capacity. Vessel feeds may be incomplete. Inventory may be unknown. A media spike may be wrong. The response team still needs a defensible next move:

- + What do we know right now?
- + What is inferred, modeled or assumed?
- + What happens if no one acts?
- + Which alternatives survive hard constraints?
- + Who is accountable for the final decision?

THE LINE I WOULD NOT CROSS

Sanjiv recommends; a human decides. It can observe, freeze a scenario, simulate consequences, compare candidate plans, audit the evidence and preserve a decision record.

It cannot place an order, charter a tanker, release a reserve, control a pipeline or operate a refinery. Public capacity metadata does not establish current fill, cargo ownership or commercial availability. That boundary is deliberate. [1][2]

PRIMARY USERS

01
Maritime and geopolitical analyst

Vessel and chokepoint activity, source health, risk explanation, provenance and historical comparison.

02
Refinery procurement head

Candidate suppliers, grades, landed cost, arrival timing, compatibility, sanctions and corridor risk.

03
Strategic reserve planner

Capacity versus opening-fill truth, draw schedule, logistics, policy floors and replenishment constraints.

04
Reviewer and approver

Evidence coverage, recomputation, rejected alternatives, immutable comments and explicit authority.

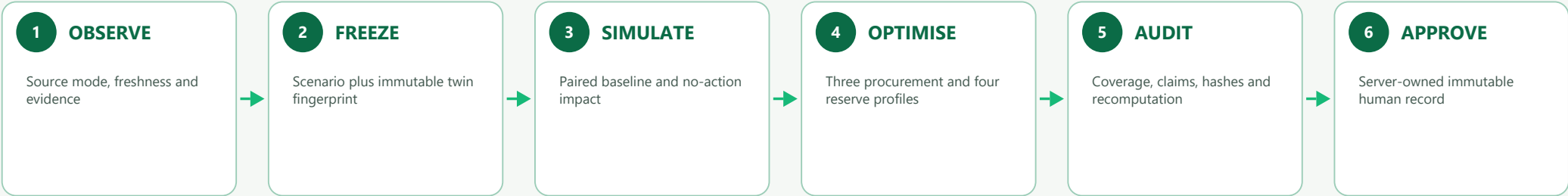
SCOPE DELIVERED IN THE REPOSITORY

Commodity	Operating mode	Decision outputs	Excluded claims
Crude oil; typed LPG extension	Live-fetched PortWatch observation plus checksummed replay and fixtures	No-action simulation; procurement; reserve; audit; approval; replay export	No live AIS claim; no private inventory; no confirmed cargo or contract; no execution

Follow one decision from signal to sign-off

Every handoff leaves a fingerprint. If evidence disappears along the way, the decision metric disappears with it.

THE SIX HANDOFFS



WHAT SITS BEHIND THE SCREENS

COMMAND CENTER	Next.js / React / typed REST and WebSocket clients
DOMAIN CORE	FastAPI modular monolith: sources, maritime, twin, scenario, simulation, optimisation, risk, audit, replay
WORKERS	Ingestion, scheduled refresh and compute workers import the same domain services
STATE	PostgreSQL + PostGIS + TimescaleDB (authoritative) Redis (cache/fan-out) MinIO (immutable artifacts)

I kept the first deployment deliberately compact: one measured modular monolith with shared domain services. Multiple API replicas come only after Redis WebSocket fan-out and deployment identity are in place.

FIVE WORDS WITH STRICT MEANINGS

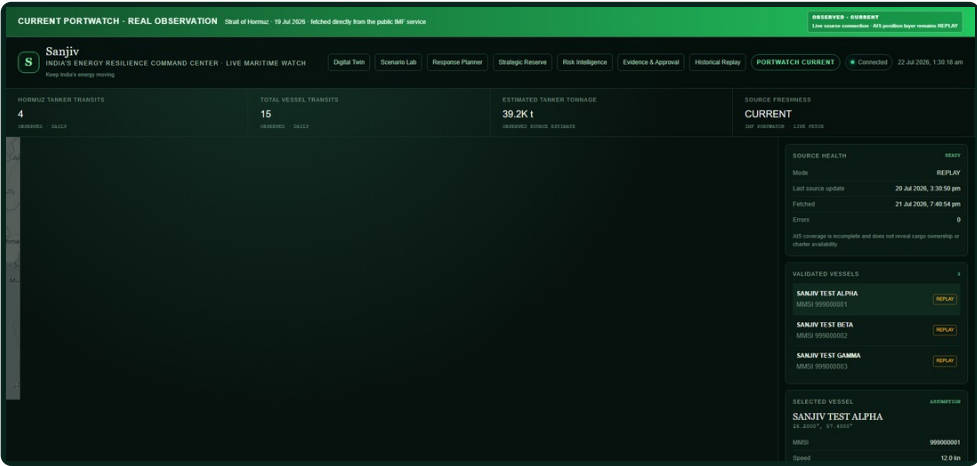
OBSERVED	Directly retrieved or supplied; normalization only.
DERIVED	Deterministic calculation from identified inputs.
INFERRED	Heuristic or probabilistic classification.
MODELED	Simulator or optimiser output.
ASSUMPTION	Visible editable input used when verification is absent.

Required fields: source references; effective, fetch and compute timestamps; freshness; confidence; evidence IDs; transformation; units; model and contract versions.

Real observations and replay stay visibly separate

PortWatch is live-fetched public data marked OBSERVED and CURRENT. The AIS position layer and modeled risk remain separately labeled replay or fixture evidence.

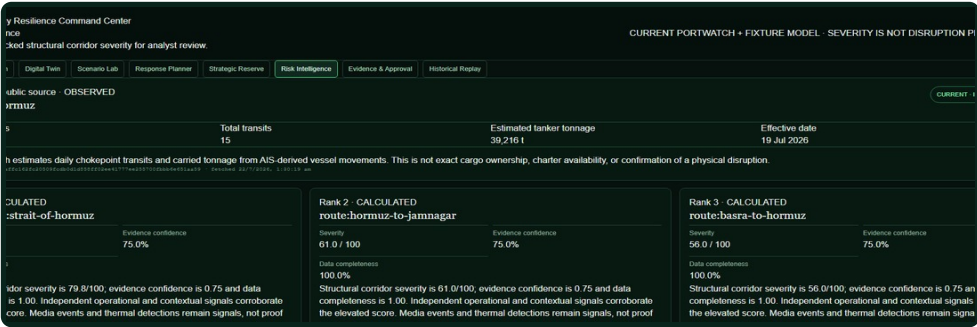
OBSERVE - COMMAND CENTER



The public IMF PortWatch service returned 4 tanker transits, 15 total transits and 39,216 t estimated tanker tonnage for 2026-07-19. The fetch is OBSERVED / CURRENT; it is a daily AIS-derived estimate, not live vessel tracking.

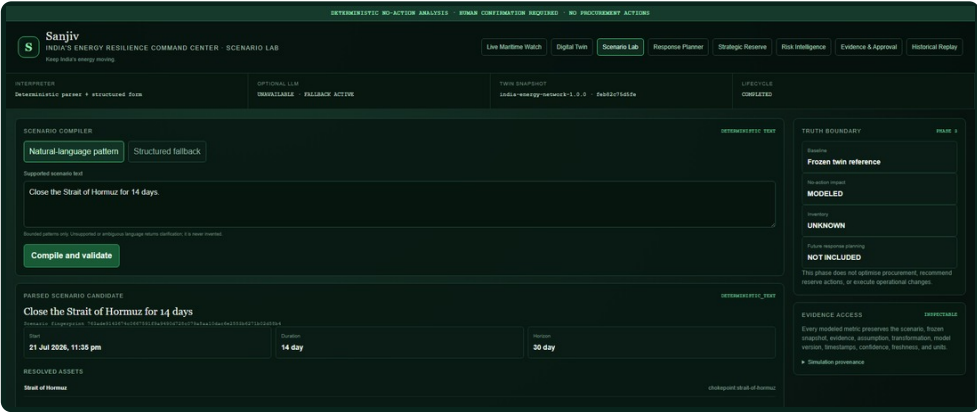
Source plane	Truth shown in the application
PortWatch daily passage	LIVE fetch; OBSERVED; CURRENT; source date and fetch time visible
AIS vessel positions	REPLAY; synthetic vessels; reported and inferred fields remain distinct
Risk model	FIXTURE / replay inputs remain separate from the observed PortWatch card

RISK INTELLIGENCE - OBSERVED SIGNAL



The observed PortWatch card sits above the structural risk ranking. The model remains explicitly fixture-based and severity is not presented as disruption probability.

SIMULATE - SCENARIO LAB



A person confirms the parsed disruption before the twin is frozen. Baseline and no-action outputs share one fingerprint; missing inventory remains UNKNOWN.

Give the human real choices, not one magic answer.

Each candidate exposes the trade-off it is making. Public reserve capacity never quietly becomes an invented opening fill.

OPTIMISE - RESPONSE PLANNER

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Response Planner

Checked procurement alternatives for modeled shortfall. Commercial availability remains assumption-dependent.

MODELED - SYNTHETIC COMMERCIAL FIXTURE

Live Maritime Watch

Digital Twin

Scenario Lab

Response Planner

Strategic Reserve

Risk Intelligence

Evidence & Approval

Historical Replay

Generate all three profiles

Completed scenario run ID

e97471a-4b41-5947-8230-c900bbe8bcb9

Generate modeled plans

Sanjiv recommends planning guidance only. It does not purchase cargo, book tankers, confirm commercial availability, or release reserves.

MODELED recommendation

LOWEST COST

Solver

OPTIMAL

Checker

PASSED

Runtime

0.0221 second

Delivered

3750 000 ktonne

Shortage

2250 000 ktonne

Objectives

2252314 229 objective_point

Supplier peak 35.0% corridor peak 35.0%

MODELED recommendation

BALANCED

Solver

OPTIMAL

Checker

PASSED

Runtime

0.0066 second

Delivered

3750 000 ktonne

Shortage

2250 000 ktonne

Objectives

2251877 100 objective_point

Supplier peak 35.0% corridor peak 35.0%

MODELED recommendation

HIGHEST RESILIENCE

Solver

OPTIMAL

Checker

PASSED

Runtime

0.0066 second

Delivered

3750 000 ktonne

Shortage

2250 000 ktonne

Objectives

2250789 609 objective_point

Supplier peak 35.0% corridor peak 35.0%

Supplier, grade, refinery and route allocations

LOWEST_COST

1650 ktonne - supplier ac7c1a6c - grade 00519e7 - refinery a0b5b75 - route 93b6a262

2100 ktonne - supplier e68b860 - grade 7d01c477 - refinery c3d93bf - route 3c308673

BALANCED

1650 ktonne - supplier ac7c1a6c - grade 00519e7 - refinery a0b5b75 - route 93b6a262

Route map and delivery timeline

Supplier → load port → corridor → receiving port → refinery

21/7/2026, 11:35:41 pm - 20/9/2026, 11:35:41 pm

21/7/2026, 11:35:41 pm - 26/7/2026, 5:35:41 pm

21/7/2026, 11:35:41 pm - 20/9/2026, 11:35:41 pm

21/7/2026, 11:35:41 pm - 26/7/2026, 5:35:41 pm

21/7/2026, 11:35:41 pm - 20/9/2026, 11:35:41 pm

21/7/2026, 11:35:41 pm - 26/7/2026, 5:35:41 pm

Objective contributions

LOWEST_COST

landed_cost: 2309 2500 objective_point

shortfall: 2250000 0000 objective_point

delay: 3.0356 objective_point

route_risk: 0.3383 objective_point

supplier_concentration: 0.7000 objective_point

corridor_concentration: 0.7000 objective_point

compatibility_penalty: 0.0000 objective_point

emissions: 0.2670 objective_point

BALANCED

landed_cost: 1847 4000 objective_point

shortfall: 2250000 0000 objective_point

delay: 12.1429 objective_point

route_risk: 2.5275 objective_point

supplier_concentration: 7.0000 objective_point

corridor_concentration: 7.0000 objective_point

compatibility_penalty: 0.0000 objective_point

emissions: 1.0350 objective_point

HIGHEST_RESILIENCE

landed_cost: 692 7750 objective_point

shortfall: 2250000 0000 objective_point

delay: 12.1429 objective_point

route_risk: 2.5275 objective_point

supplier_concentration: 7.0000 objective_point

corridor_concentration: 7.0000 objective_point

compatibility_penalty: 0.0000 objective_point

emissions: 1.0350 objective_point

Builder's note. LOWEST COST, BALANCED and HIGHEST RESILIENCE share the same hard constraints and input fingerprint. The point is not a magic answer; it is a legible choice with checker state, shortage, cost components, rejections, evidence and assumptions.

RESERVE - STRATEGIC RESERVE

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Strategic Reserve

Site-level continuity guidance coordinated with one exact checked procurement plan.

MODELED - ASSUMPTION-DEPENDENT OPENING FILL

Live Maritime Watch

Digital Twin

Scenario Lab

Response Planner

Strategic Reserve

Risk Intelligence

Evidence & Approval

Historical Replay

Generate policy guidance

Completed scenario run ID

e97471a-4b41-5947-8230-c900bbe8bcb9

Checked procurement plan ID

eca30e6b-4c0e-5400-b016-5ae52448006

Generate modeled guidance

Sanjiv recommends guidance only. It does not release reserves, purchase cargo, operate pipelines, or escalate replenishment.

MODELED recommendation

CONSERVATIVE

Solver

OPTIMAL

Checker

PASSED

Runtime

0.0050 second

Residual shortage

2250 000 ktonne

MODELED recommendation

BALANCED

Solver

OPTIMAL

Checker

PASSED

Runtime

0.0049 second

Residual shortage

2065 000 ktonne

MODELED recommendation

AGGRESSIVE CONTINUITY

Solver

OPTIMAL

Checker

PASSED

Runtime

0.0040 second

Residual shortage

1950 000 ktonne

MODELED recommendation

NO RESERVE USE

Solver

OPTIMAL

Checker

PASSED

Runtime

0.0041 second

Residual shortage

2250 000 ktonne

Three reserve sites: capacity and opening-fill truth

CONSERVATIVE

Mangalore Strategic Reserve: capacity 1500 000 ktonne (OBSERVED) - opening 980 000 ktonne (UNEXPIRED_ASSUMPTION) - floor 980 000 ktonne

Visakhapatnam Strategic Reserve: capacity 1030 000 ktonne (OBSERVED) - opening 720 000 ktonne (UNEXPIRED_ASSUMPTION) - floor 720 000 ktonne

Padur Strategic Reserve: capacity 2500 000 ktonne (OBSERVED) - opening 1650 000 ktonne (UNEXPIRED_ASSUMPTION) - floor 1650 000 ktonne

BALANCED

Mangalore Strategic Reserve: capacity 1500 000 ktonne (OBSERVED) - opening 980 000 ktonne (UNEXPIRED_ASSUMPTION) - floor 825 000 ktonne

Visakhapatnam Strategic Reserve: capacity 1030 000 ktonne (OBSERVED) - opening 720 000 ktonne (UNEXPIRED_ASSUMPTION) - floor 569 500 ktonne

Padur Strategic Reserve: capacity 2500 000 ktonne (OBSERVED) - opening 1650 000 ktonne (UNEXPIRED_ASSUMPTION) - floor 875 000 ktonne

AGGRESSIVE CONTINUITY

Mangalore Strategic Reserve: capacity 1500 000 ktonne (OBSERVED) - opening 980 000 ktonne (UNEXPIRED_ASSUMPTION) - floor 525 000 ktonne

Visakhapatnam Strategic Reserve: capacity 1030 000 ktonne (OBSERVED) - opening 720 000 ktonne (UNEXPIRED_ASSUMPTION) - floor 360 500 ktonne

Padur Strategic Reserve: capacity 2500 000 ktonne (OBSERVED) - opening 1650 000 ktonne (UNEXPIRED_ASSUMPTION) - floor 875 000 ktonne

NO RESERVE USE

Mangalore Strategic Reserve: capacity 1500 000 ktonne (OBSERVED) - opening 980 000 ktonne (UNEXPIRED_ASSUMPTION) - floor 980 000 ktonne

Visakhapatnam Strategic Reserve: capacity 1030 000 ktonne (OBSERVED) - opening 720 000 ktonne (UNEXPIRED_ASSUMPTION) - floor 720 000 ktonne

Padur Strategic Reserve: capacity 2500 000 ktonne (OBSERVED) - opening 1650 000 ktonne (UNEXPIRED_ASSUMPTION) - floor 1650 000 ktonne

Release schedule, path and receiving refinery

BALANCED: 155 000 ktonne guidance - site 20b61422 → route ea690db → refinery b0c7e08 - receipt 24/7/2026, 11:35:41 am

AGGRESSIVE_CONTINUITY: 300 000 ktonne guidance - site 20b61422 → route ea690db → refinery b0c7e08 - receipt 24/7/2026, 11:35:41 am

Remaining inventory and cover

CONSERVATIVE: 980 000 ktonne - 13.067 day

CONSERVATIVE: 980 000 ktonne - 13.067 day

CONSERVATIVE: 720 000 ktonne - 720000000000 000 day

CONSERVATIVE: 720 000 ktonne - 720000000000 000 day

CONSERVATIVE: 1650 000 ktonne - 165000000000 000 day

CONSERVATIVE: 1650 000 ktonne - 165000000000 000 day

BALANCED: 980 000 ktonne - 13.067 day

BALANCED: 825 000 ktonne - 11.000 day

BALANCED: 720 000 ktonne - 720000000000 000 day

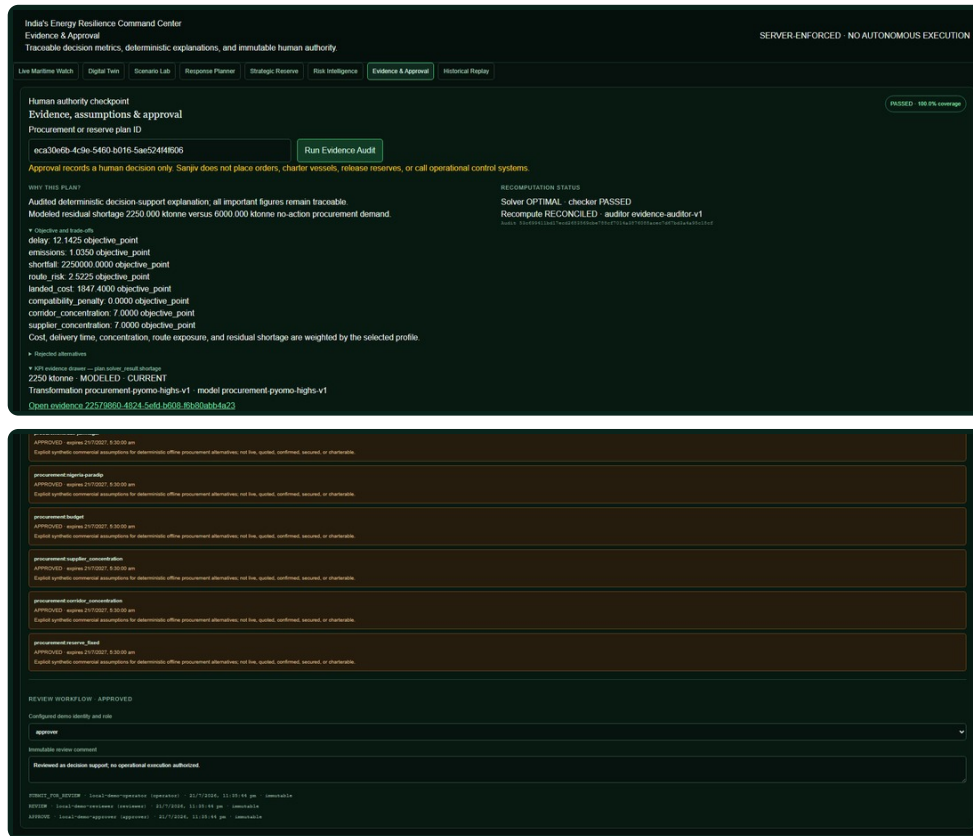
BALANCED: 720 000 ktonne - 720000000000 000 day

Builder's note. Capacity is not fill. The screen keeps public site capacity separate from the synthetic opening-fill assumption, then checks floors, draw and receipt limits, transit and residual shortage. No replenishment input means no hidden replenishment.

A plan is not trusted because the interface says 'optimal'.

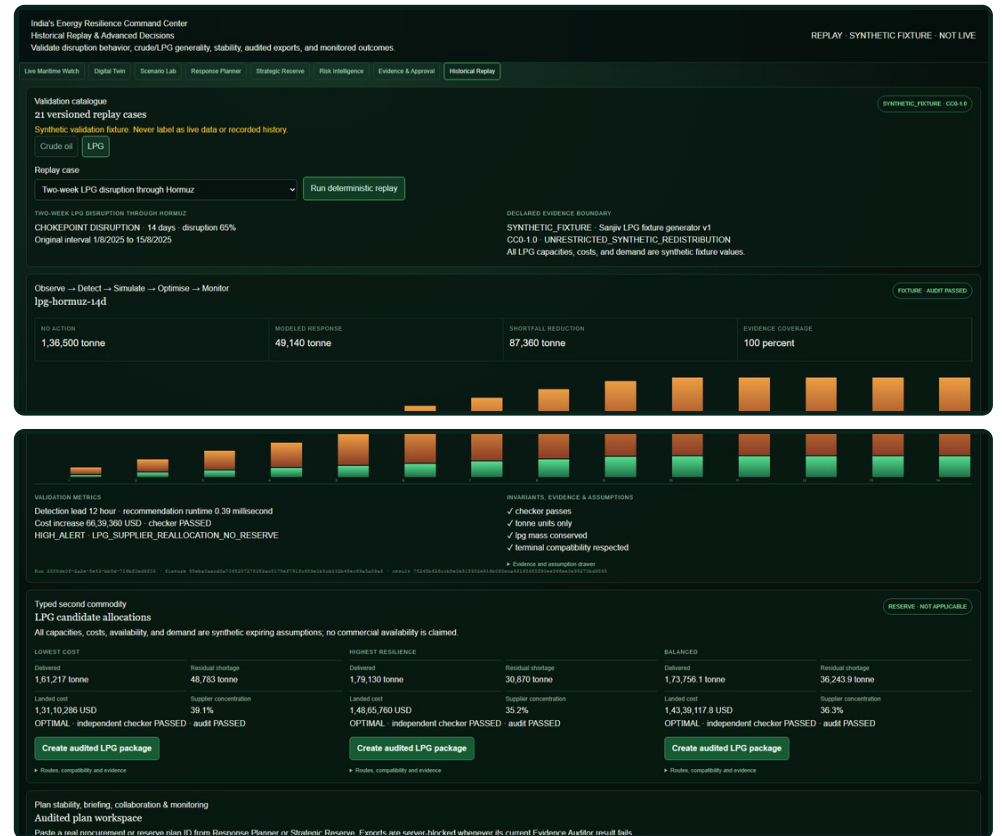
Trust comes from the evidence, the arithmetic and the person willing to sign their name to the decision.

AUDIT AND APPROVE - EVIDENCE WORKSPACE



Builder's note. The captured audit is PASSED with 100.0% coverage and RECOMPUTE_RECONCILED. The approval stores server-resolved actor, role, UTC time and an immutable comment. It records a human decision; it never executes one.

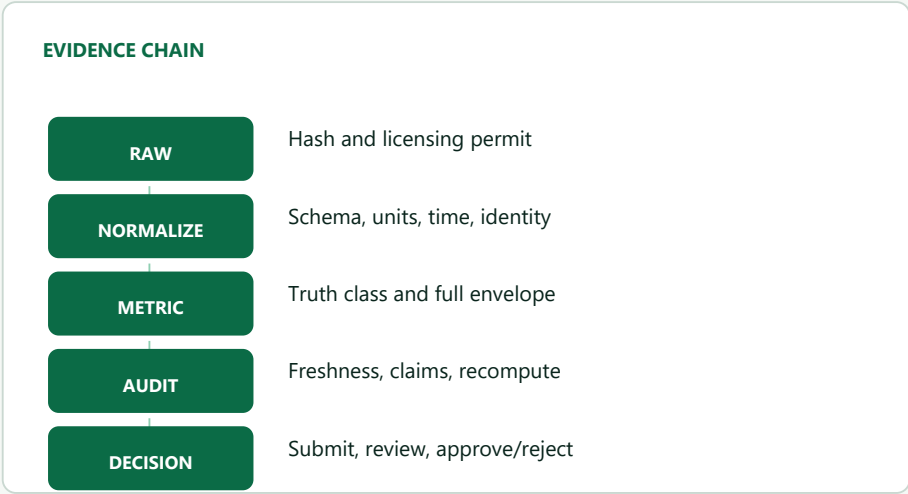
MONITOR - HISTORICAL REPLAY AND LPG



Builder's note. Twenty-one versioned CC0 synthetic cases keep regression and demonstration repeatable. Classification, generator, license, assumptions, invariants and audit state stay visible. Seeded sensitivity is deterministic analysis, not a probability forecast.

Provenance must survive a hard question

If someone asks, 'Where did that number come from?', the answer should be one click away - or the number should not be on the decision screen.



Missing mandatory evidence blocks the metric from decision UI and briefing output.

REGISTERED EXTERNAL SOURCES AND EXACT TRUTH TREATMENT

Source	Repository use	Truth and limitation	Ref.
PPAC	Typed offline import boundary for refinery and energy statistics	Bundled operating values are assumptions; no PPAC payload is bundled	[1]
ISPRL	Public reserve-site capacity and connectivity metadata	Capacity may be observed; current fill stays unknown without verified input	[2]
IMF PortWatch	Live-fetched daily passage observation and baseline	Observed public estimate; current by source cadence; not live AIS	[3]
OFAC	Official sanctions list files	List entry observed; exact match derived; fuzzy match inferred	[4]
EIA / FRED	Optional price, supply and macro series	Series cadence, units, rights and attribution remain source-specific	[5][6]
NASA FIRMS / GDELT	Optional thermal and media signals	Signal is not proof of closure, attack, cause or damage	[7][8]

SECURITY AND AUTHORITY BOUNDARY

+ Credentials stay server-side; browser configuration contains only the public API origin.

+ Production API and governance fail closed without configured server-side identities.

+ Origin, request size/type, rate, SSRF-safe adapter and bounded solver controls are implemented.

+ Independent checkers and the Evidence Auditor block failed plans, approvals and exports.

+ Terminal plans, audits, approvals, exports and snapshots are append-only or database-immutable.

Residual boundary: deployment IdP, TLS termination, multi-user session policy and Redis fan-out are operator-owned prerequisites.

08 / VALIDATION

Here is what actually passed - and what it does not prove

Machine, dataset, timestamp and commit matter. Current tests were rerun; performance evidence was regenerated from a clean committed application source.

CURRENT NPM TEST

212 passed

188 Python + 23 web + 1 contracts

FAILED / SKIPPED

0 / 0

Exit 0; elapsed 146.6 s

BROWSER E2E COVERAGE

2 journeys

All 8 product screens; Observe to Monitor

REPLAY CATALOGUE

21 cases

Checksummed CC0 synthetic fixture

LOCAL FIXTURE PERFORMANCE (MILLISECONDS)

Metric	Median	p95	n
AIS ingest to map	25.214	25.214	1
Risk scoring	36.928	48.208	5
Scenario compilation	9.541	14.313	5
Simulation	63.616	83.267	5
Procurement optimisation	260.506	541.371	5
Reserve optimisation	180.672	220.694	5
Evidence audit	53.661	61.946	5
Briefing export	129.634	296.531	5
Signal to recommendation	712.521	1062.676	5

Report classification: MEASURED_LOCAL_SYNTHETIC_FIXTURE. Five samples unless shown otherwise. These are release measurements, not production SLAs or model-accuracy claims.

BENCHMARK IDENTITY

Run 54f808c9-c228-41cc-9662-8af1479295a6
Report commit 14a7e5deb639aeb6de4e683ec62cd22abbea0838
Source state CLEAN_COMMITTED; timestamp 2026-07-21T20:06:46.912821Z

BROWSER, LOAD AND RESYNC

Map 42.85 FPS; interaction 800.482 ms; frame-time p95 57.2 ms across 120 frames.
Load: 64/64 successful at concurrency 8; p95 428.578 ms. Resync observed after 100 publishes; p95 1.133 ms.

DEPENDENCY FAILURE AND RECOVERY DRILL (MS)

Component	Detection	Recovery	Result
Redis	1030.028	30.933	PASS
MinIO	1028.974	1487.670	PASS
PostgreSQL	1028.329	1035.884	PASS
Compute worker	22427.376	1045.361	PASS
API	n/a	4754.407	PASS

Security: PASS (8 checks; 0 failure names). Backup/restore: PASS; 2,673,646 bytes; migration 20260721_0009; corrupt artifact rejected.

09 / PATH TO USE

The value is a better decision record

The repository proves observed-source ingestion, workflow mechanics and deterministic checks on fixtures. Operational value still requires licensed history and verified operator inputs.

DECISION VALUE NOW

- + One immutable chain from source state to approval.
- + Paired no-action and response alternatives on the same frozen inputs.
- + Independent feasibility checks and named rejected options.
- + Visible assumption ownership, expiry, freshness and evidence coverage.
- + Audited JSON/PDF packages plus monitored replay outcomes.

No rupee savings, avoided-loss value or decision-time reduction was measured.

VERIFIED DEPLOYMENT SHAPE

Docker Compose runs the same web/API images, three workers, PostgreSQL with PostGIS and TimescaleDB, Redis and MinIO for local and demo use.

- + Liveness and dependency-aware readiness
- + Source, worker and request-runtime status
- + Structured logs and trace-compatible identifiers
- + Backup/restore and single-dependency drills

SCALING POLICY

- + Scale vertically first using measured queue and request latency.
- + Add ingestion/refresh/compute worker replicas by workload.
- + Add Redis WebSocket fan-out before multiple API replicas.
- + Use managed database/object storage, external TLS and an operator collector.
- + Require an ADR and workload evidence before Kubernetes or domain microservices.

OPERATIONAL ADOPTION ROADMAP

01

Replace fixtures

Licensed recorded history and verified private/operator inputs.

02

Calibrate

Out-of-sample thresholds, model weights and decision policies with domain owners.

03

Secure identities

Deployment IdP, TLS edge, rotation, sessions and least privilege.

04

Scale and recover

Redis fan-out, managed HA, encrypted off-site backup and multi-region drills.

05

Accept and govern

Operator acceptance, data rights, model monitoring and change control.

METRICS FOR A REAL DEPLOYMENT - NOT RESULTS IN THIS SUBMISSION

Shortfall reduction; incremental cost per avoided shortage unit; inventory-cover extension; refinery-utilisation recovery; supplier and corridor concentration; decision time saved. Each requires a baseline, verified input data and an agreed measurement window.

10 / LIMITS AND REFERENCES

What I refused to fake

A convincing demo should make its limits easier to see, not easier to miss. Real PortWatch observations and fixture-based decision evidence remain explicitly separated.

CURRENT LIMITATIONS

- PortWatch passage data is live-fetched and labeled OBSERVED / CURRENT; AIS positions, decision-model inputs and the replay catalogue remain fixture or replay evidence where shown.
- Commercial price, supplier availability, port/refinery values and reserve opening fill are synthetic assumptions unless replaced by verified operator records.
- No order placement, tanker charter, reserve release, pipeline or refinery-control integration exists.
- Risk weights, alert thresholds and deterministic sensitivity are not calibrated disruption probabilities.
- The benchmark is a one-machine local synthetic-fixture run from clean application commit 14a7e5deb639; it is release evidence, not a production SLA or model-accuracy result.
- The recovery drill stops one dependency at a time on a local single-host stack; it does not establish multi-region failover.
- Production requires deployment IdP, TLS, key rotation, sessions, managed backups and Redis fan-out before horizontal API scaling.
- Licensed recorded validation, domain-owner calibration and operator acceptance remain required before operational decision use.

AUTHORITATIVE REFERENCES

[1] PPAC - Installed Refining Capacity

<https://ppac.gov.in/infrastructure/installed-refinery-capacity>

[2] ISPRIL - Strategic reserve facilities

<https://isprilindia.com/aboutus.asp>

[3] IMF PortWatch - Data and methodology

<https://portwatch.imf.org/pages/data-and-methodology/>

[4] OFAC - Sanctions List Service

<https://ofac.treasury.gov/sanctions-list-service>

[5] U.S. EIA - API v2 documentation

<https://www.eia.gov/opendata/documentation.php>

[6] FRED - API overview

<https://fred.stlouisfed.org/docs/api/fred/overview.html>

[7] NASA FIRMS - Area API

<https://firms.modaps.eosdis.nasa.gov/api/area/>

[8] GDELT - DOC 2.0 API

<https://blog.gdeltproject.org/gdelt-doc-2-0-api-debuts/>

SUBMISSION LINKS



Repository QR

Repository

<https://github.com/vivekyarra/Sanjiv>

Demo

<https://drive.google.com/drive/folders/1JDFeLyQOd4g74BhE63Yuh6OCGmCoaf6K?usp=sharing>

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Application commit

14a7e5deb639aeb6de4e683ec62cd22abbea0838